

Sensor-Family Composition of the In-Situ Archive Changes the Apparent Bias and RMSD of Satellite Soil-Moisture Products

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Abstract—Global satellite and reanalysis soil-moisture products are commonly validated against the International Soil Moisture Network (ISMN), whose reference-error structure is often treated as independent of sensor manufacturer. This letter quantifies how that composition assumption affects reported product metrics. Across 1,625 ISMN station-sensors (2020–2023), 78.4 % belong to one dielectric family (Stevens HydraProbe), whose mean residual is drier than the references while Decagon/METER TDR records are wetter—a $0.17 \text{ m}^3 \text{ m}^{-3}$ family swing that exceeds the ERA5-Land/SMAP L3_E/SMAP-SMI disagreement. Propagated into product metrics, the mean-state bias correction inferred from ISMN changes by a factor of five for ERA5-Land and changes sign for SMAP L3_E depending on which sensor family is used, and the SMAP-SMI mean-state RMSD increases by 41 % on the non-HydraProbe subsample. A geography-controlled paired-sensor test and a dielectric-independent cosmic-ray-neutron cross-check, including a USA-excluded sensitivity, indicate that the residual structure is associated with sensor family after geography and network checks. A 2021 overpass-matched SMAP temporal check also shows family structure in ubRMSE and temporal correlation metrics. ISMN-based validation metrics should be reported with sensor-family stratification or reweighting.

Index Terms—Soil moisture, validation, SMAP, ERA5-Land, ISMN, in-situ reference, sensor bias, cosmic-ray neutron sensor.

I. INTRODUCTION

THE quantitative skill of every global satellite soil-moisture retrieval and reanalysis product is reported as a validation statistic—a network-mean bias, RMSD, or correlation—computed against in-situ observations. In practice this reference is supplied almost exclusively by the International Soil Moisture Network (ISMN) [1], which aggregates more than twenty national and regional networks into a common reference archive and is used downstream to quote a single number per candidate product, including for SMAP [4], [5] and ERA5-Land [3]. Community validation protocols emphasize site representativeness, error characterization, and

transparent reporting of bias, ubRMSE, and correlation metrics [6], [2].

Aggregating the archive this way implicitly assumes that its error model is *manufacturer-neutral*: that the reference is instrument-blind, so a product can be scored without regard to the particular sensors that happen to populate the archive. That assumption need not hold. Dielectric soil-moisture sensors from different vendors operate at different frequencies (e.g., 50 MHz for the Stevens HydraProbe [10]) and use different calibration curves. If the archive is dominated by one family and that family carries a systematic near-surface offset, the offset does not average out—it is inherited by the validation benchmark of every product scored against the archive.

This letter evaluates that assumption as a remote-sensing validation question: *how much do the reported bias and RMSD of ERA5-Land, SMAP L3_E, and the SMAP-derived SMI product change when the ISMN reference is stratified by sensor-manufacturer family?* The analysis quantifies the product-level sensitivity directly, controls for deployment geography with co-located sensor pairs and network-clustered inference, and adds a dielectric-independent cosmic-ray-neutron (CRNS) cross-check. The inference is not a laboratory calibration claim; it is a validation-methodology result showing that the ISMN benchmark is not sensor-family neutral for these mean-state product metrics.

II. DATA AND METHODS

A. ISMN extraction and family assignment

All *_sm*.stm files in the ISMN archive used here (31,201 files, 2015–2023) were parsed; we retained records dated entirely in 2020–2023 with nominal depth from $\leq 0.15 \text{ m}$ and > 180 valid hourly observations per year, giving 1,625 unique (network, station, sensor, depth) records across 23 networks on all continents. Each sensor string was matched to a manufacturer family (HydraProbe family, Decagon/METER TDR, Campbell CS6xx, ThetaProbe, TRIME, other); strings containing *hydraprobe* or *stevens* form one HydraProbe family (the same Stevens Water Monitoring 50 MHz probe under several model numbers). The full composition is given in Supplement Table S1.

B. Reference products and processing

For each product we sampled each station by nearest valid cell and formed the residual (ISMN – reference). ERA5-Land [3]: layer-1 (0–7 cm) volumetric soil water, per-cell

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Processing scripts and derived station-reference and CRNS-ISMN matchup summary tables are archived on Zenodo (doi:10.5281/zenodo.21263063) and mirrored in the public source-code repository (v20260708): https://github.com/ecomm77/soil_moisture_organic_bias.

temporal mean over 2020–2023 at 0.1° . *SMAP L3_E* [4]: SPL3SMP_E v006; the analysis used 182 available days from 2023, combining ascending and descending passes and retaining only recommended-quality retrievals (retrieval-quality flag) into a 9-km EASE-Grid mean field, with a 0.15° cell rejection radius. *SMAP-SMI* [9]: 0.1° product, native 2020–2023, on a stratified 192-day sample. A temporal-window control (2020–2023 vs. 2023-only ERA5-Land) leaves the family ordering and the HydraProbe-family dry residual unchanged (family change $\leq 0.012 \text{ m}^3 \text{ m}^{-3}$; Supplement).

Scope of the metrics. These are *mean-state* validation metrics—the bias and RMSD of multi-year mean fields—not overpass-matched ubRMSE or anomaly correlation [7], [8]. This is the quantity represented by a single network-mean number per product. To connect the result to standard Cal/Val language, we also report a 2021 overpass-matched SMAP L3 SM_P temporal check in the Supplement.

C. Attribution and confounding controls

Because HydraProbe-heavy networks (SCAN, SNOTEL, USCRN) also occupy particular climates and soils, sensor family and deployment geography are correlated. We separate them three ways. (i) *Co-located pairs*: stations hosting both a HydraProbe-family and a non-HydraProbe sensor in the same ISMN grid cell ($n = 21$), compared with a paired *t*-test; both members lie in the retained near-surface layer (depth-from $\leq 0.15 \text{ m}$), and the corresponding depth constraint is described in the Supplement. (ii) *Network-clustered inference*: a cluster bootstrap (5,000 resamples) that resamples ISMN networks as the sampling unit, plus leave-one-network deletion, used as the primary uncertainty rather than naïve per-station bootstrap. (iii) *Dielectric-independent cross-check*: because product residuals alone cannot separate an in-situ sensor effect from product-specific error, we add a CRNS reference whose neutron signal is independent of soil permittivity [11]. Four archives (COSMOS-USA [15], COSMOS-Europe [12], COSMOS-UK [13], CosmOz Australia [14]) were matched to ISMN within 0.1° – 0.5° . For COSMOS-USA, the CRNS–ISMN residual is assembled as the Zheng et al. CRNS–ERA5-Land bias minus the station-matched ISMN–ERA5-Land residual from our 0.1° overlay; the non-USA archives use direct CRNS–probe or local matchup contrasts. The pooled HydraProbe-minus-non-HydraProbe contrast and its clustered bootstrap are reported here, with archive-level detail and a USA-excluded sensitivity in the Supplement.

III. RESULTS

A. The validation archive is manufacturer-asymmetric

Of the 1,625 station-sensors, 1,274 (78.4%) are HydraProbe-family; all other families combined are $< 22\%$. Within the same retained archive, the family mean residual against ERA5-Land is $-0.097 \text{ m}^3 \text{ m}^{-3}$ for HydraProbe-family station-sensors but $+0.068 \text{ m}^3 \text{ m}^{-3}$ for Decagon/METER TDR (Fig. 1). Thus the validation archive is not only manufacturer-asymmetric in its counts; it also contains family-structured mean-state residuals. A single archive-mean validation number is therefore influenced by which families and sub-networks dominate the retained ISMN sample.

B. Product validation metrics depend on sensor-family composition

Stratifying the ISMN reference by family changes the validation statistics of all three products (Fig. 2). The mean residual swings from $-0.097 \text{ m}^3 \text{ m}^{-3}$ for the HydraProbe family to $+0.068 \text{ m}^3 \text{ m}^{-3}$ for Decagon/METER TDR against ERA5-Land (a $0.17 \text{ m}^3 \text{ m}^{-3}$ swing, larger than the between-product disagreement). Propagated into commonly reported validation metrics: the ISMN-implied ERA5-Land mean-state bias correction is -0.081 over the full archive, -0.097 on the HydraProbe subsample, and -0.019 on the non-HydraProbe subsample—a factor-of-five change in amplitude. For SMAP L3_E the same triple is $-0.016/-0.025/+0.008$: the inferred correction *flips sign*, so whether SMAP reads wetter or drier than the in-situ reference depends on sensor-family composition. The mean-state RMSD is likewise composition-dependent; for SMAP-SMI it rises from 0.099 (HydraProbe subsample) to $0.140 \text{ m}^3 \text{ m}^{-3}$ (non-HydraProbe subsample), a 41% inflation. Because HydraProbe dominates the archive roughly three-to-one, the full-archive number is weighted toward the HydraProbe-family subset.

Equal-weighting the HydraProbe and non-HydraProbe subsets changes the ERA5-Land mean residual from -0.081 to $-0.058 \text{ m}^3 \text{ m}^{-3}$ and the SMAP L3_E residual from -0.016 to $-0.009 \text{ m}^3 \text{ m}^{-3}$. Equal weighting across the five non-TRIME families gives $-0.038 \text{ m}^3 \text{ m}^{-3}$ for ERA5-Land and $+0.001 \text{ m}^3 \text{ m}^{-3}$ for SMAP L3_E, illustrating how a simple reweighting protocol converts the diagnostic into an actionable reporting rule. The two primary independent checks remain significant after Holm correction ($p_{\text{Holm}} = 4.6 \times 10^{-4}$ for CRNS and 4.7×10^{-3} for co-located pairs).

C. Overpass-matched temporal check

A real 2021 SMAP L3 SM_P (AM) overpass-matched check adds the Cal/Val metrics that the mean-state analysis intentionally does not target (Supplement Table S3, Fig. S2). Across 785 station-sensors and 78,600 matched station-days, the pooled HydraProbe versus non-HydraProbe bias contrast is small but significant (-0.032 versus $-0.036 \text{ m}^3 \text{ m}^{-3}$; $p = 9.8 \times 10^{-3}$) because the non-HydraProbe bin mixes wetter Decagon/METER TDR records ($+0.058 \text{ m}^3 \text{ m}^{-3}$) with drier Campbell and other records (-0.107 and $-0.071 \text{ m}^3 \text{ m}^{-3}$). The headline family pair therefore remains aligned with the mean-state result: HydraProbe is drier than Decagon/METER by $0.090 \text{ m}^3 \text{ m}^{-3}$, and Decagon/METER also has higher Pearson *R* (0.622 versus 0.317) and anomaly *R* (0.423 versus 0.329). Pooled ubRMSE differs (0.062 versus $0.057 \text{ m}^3 \text{ m}^{-3}$; $p = 8.2 \times 10^{-4}$), while pooled Pearson *R* does not ($p = 0.70$) and pooled anomaly *R* is marginal ($p = 0.055$). This single-year diagnostic does not replace a full multi-product temporal validation, but it shows that family structure is visible in standard satellite-validation metrics as well as in mean-state bias/RMSD.

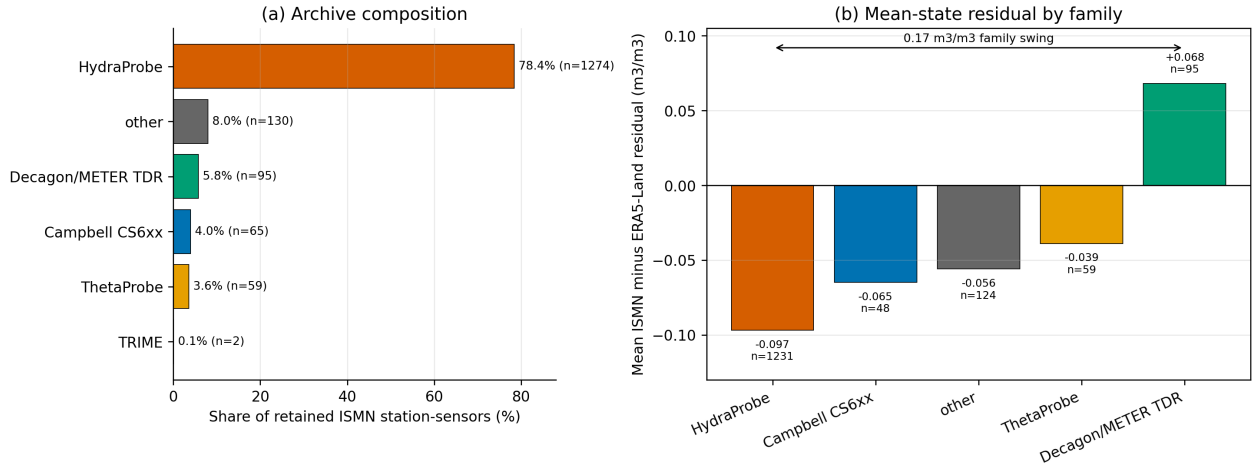


Fig. 1. The ISMN validation archive is manufacturer-asymmetric. (a) Retained 2020–2023 near-surface ISMN station-sensors by manufacturer family. HydraProbe-family records supply 78.4% of the archive. (b) Mean ISMN–ERA5–Land residual by family; residual-panel counts reflect valid ERA5–Land overlay coverage, and TRIME is omitted because $n = 2$. The HydraProbe-to-Decagon/METER contrast is $0.17 \text{ m}^3 \text{ m}^{-3}$, so archive composition can change a reported product bias even before any retrieval algorithm changes.

D. The family residual is robust to geography and network checks

Two controls address whether the effect reflects deployment geography. At the $n = 21$ co-located pairs—which hold location, climate, season, and soil constant—the non-HydraProbe sensor reads wetter by $+0.075 \text{ m}^3 \text{ m}^{-3}$ (paired $t = +3.19$, $p = 4.7 \times 10^{-3}$; CI $[+0.031, +0.119]$). Under network-clustered resampling the HydraProbe-family residual remains negative and its CI excludes zero, and every leave-one-network deletion preserves the effect ($p < 10^{-3}$). The supported interpretation is therefore limited but specific: the residuals remain sensor-family structured after network-level robustness checks.

E. Dielectric-independent CRNS cross-check

Pooling the 92 CRNS↔ISMN pairs that permit a within-archive family contrast (58 HydraProbe, 34 non-HydraProbe), the dielectric-free neutron reference sits closer to the non-HydraProbe families: mean HydraProbe residual $+0.111$ versus non-HydraProbe $-0.044 \text{ m}^3 \text{ m}^{-3}$ (Welch's $t = +4.07$, $p = 2.3 \times 10^{-4}$). The contrast survives every leave-one-archive deletion ($p < 10^{-3}$), every leave-one-network deletion ($p < 10^{-3}$), and a cluster bootstrap whose CIs exclude zero under network, CRNS-site, and archive clustering. The COSMOS-USA component is the synthesized CRNS–ERA5 minus ISMN–ERA5 residual described above; when COSMOS-USA is removed, the remaining archives still preserve the contrast ($t = +3.91$, $p = 6.3 \times 10^{-4}$). The Australian archive ($n = 5$) is directionally consistent but too small for an independent test. Per-archive results are given in Supplement Fig. S1; the cross-check adds a permittivity-independent line of evidence for the same family ordering seen in Fig. 2.

IV. DISCUSSION AND CONCLUSION

The global ISMN benchmark is dominated by a single dielectric family, and the in-situ–reference residual is sensor-

family structured. The consequence for remote sensing is direct and does not require resolving whether the offset is a calibration effect or a deployment effect: because the HydraProbe-family dry residual is not distributed symmetrically across the archive, the mean-state bias and RMSD reported for ERA5–Land, SMAP L3_E, and SMAP-SMI depend on which sensor family populates the reference—to the point of reversing the sign of an inferred SMAP correction. Manufacturer neutrality of the ISMN benchmark is therefore not a safe assumption for these mean-state metrics.

These results support treating sensor family as an explicit stratification, fixed-effect, or reweighting term in ISMN-based validation, with global product skill reported by family or on a family-balanced subsample; even equal HydraProbe/non-HydraProbe weighting materially changes the reported ERA5–Land and SMAP L3_E mean residuals. Two limitations bound the claim. First, the primary metrics are mean-state summaries, although a real 2021 SMAP temporal check shows family structure in ubRMSE and anomaly R as well. Second, the attribution, although supported by a geography-controlled pair test and a dielectric-independent CRNS cross-check, remains a residual-structure result rather than a laboratory calibration proof. A sensitivity analysis controlling for SoilGrids 2.0 clay fraction and bulk density [16] yields the same family ordering, indicating that the effect is not explained by soil texture alone.

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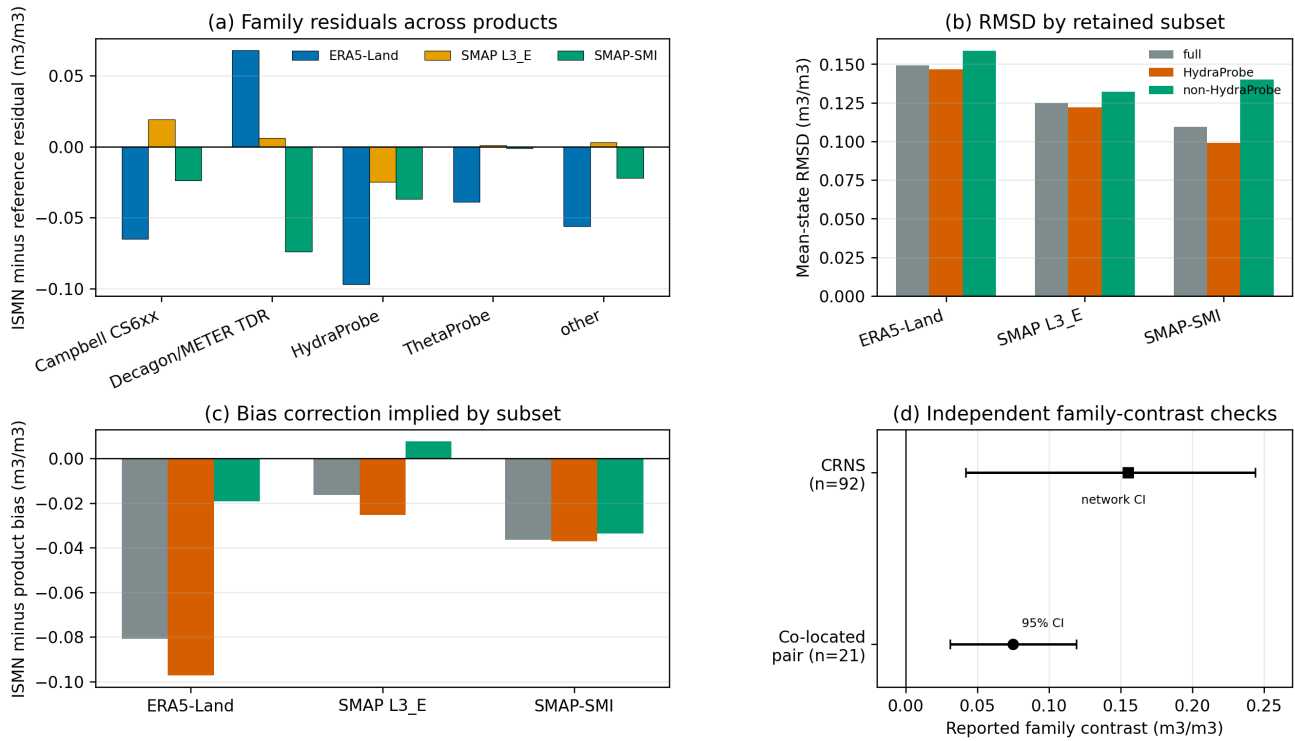


Fig. 2. Sensor-family composition changes satellite-product validation metrics. (a) Mean-state ISMN—reference residual by family for ERA5-Land, SMAP L3_E, and SMAP-SMI; the HydraProbe—Decagon/METER swing ($\approx 0.17 \text{ m}^3 \text{ m}^{-3}$) exceeds the between-product disagreement. (b) Mean-state RMSD for each product under the full archive and the HydraProbe-only and non-HydraProbe-only subsamples; the SMAP-SMI RMSD inflates 41 % on the non-HydraProbe subsample. (c) Implied ISMN—reference bias correction by subsample; the ERA5-Land amplitude changes by $\sim 5\times$ and the SMAP L3_E correction changes sign. (d) Independent family-contrast checks: co-located pairs give a non-HydraProbe-minus-HydraProbe difference of $+0.075 \text{ m}^3 \text{ m}^{-3}$ [$+0.031, +0.119$], and the CRNS cross-check gives a HydraProbe-minus-non-HydraProbe contrast of $+0.155 \text{ m}^3 \text{ m}^{-3}$ [$+0.042, +0.244$] under network-cluster bootstrap; site- and archive-cluster CIs also exclude zero. This is the practical consequence for satellite soil-moisture validation and is independent of the sensor-versus-deployment attribution.

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